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Airborne Remote Sensing of Concurrent Submesoscale Dynamics and Phytoplankton

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Key Points:

- Concurrent, airborne sub-kilometer ocean currents, sea surface temperature, and calibrated ocean color proxies are merged for the first time
- Airborne snapshots capture the impact of submesoscale dynamics on phytoplankton without spatiotemporal aliasing
- This study works toward the detection and quantification of submesoscale bio-physical mechanisms using remote sensing

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract

Submesoscale dynamics can induce significant vertical fluxes of phytoplankton, nutrients, and carbon, resulting in biological and climatological impacts such as enhanced phytoplankton production, phytoplankton community shifts, and carbon export. However, resolving these dynamics is challenging due to their rapid evolution (hours to days) and small spatial scales (1–10 km) of variability. The Modular Aerial Sensing System (MASS), an airborne instrument package measuring concurrent ocean dynamics and hyperspectral ocean color, provides a powerful tool to study the influence of submesoscale dynamics on phytoplankton and carbon. In this study, we present the first airborne observations pairing snapshots of sub-kilometer ocean velocities and their derivatives (i.e., vorticity, divergence, and strain) with concurrent ocean color and sea surface temperature. We developed airborne proxies of chlorophyll-a and particulate organic carbon, which explained about 66.2% and 56.2% of in situ variability without atmospheric correction, suggesting that MASS can capture phytoplankton variability. We also explored relationships between concurrent vorticity, divergence, strain, sea surface temperature, chlorophyll-a, and hyperspectral variables to illuminate the submesoscale processes that alter phytoplankton distributions. This study demonstrates the value of merging bio-optical and physical airborne remote sensing data to better understand the influence of submesoscale dynamics on oceanic ecosystems and organic carbon. We highlight the potential for suborbital remote sensing for studying processes that impact phytoplankton ecosystems and carbon transport without the spatiotemporal aliasing affecting in situ sensors.

Plain Language Summary

Oceanic processes that act on spatial scales of 1–10 km impact phytoplankton, with potential implications for the marine food web and ocean carbon cycle. Disentangling the impacts of submesoscale processes on phytoplankton and carbon is challenging due to their small and fast spatial and temporal scales. The Modular Aerial Sensing System (MASS) is an airborne platform capturing simultaneous measurements of submesoscale ocean dynamics and ocean color, from which phytoplankton and organic carbon distributions can be derived. The relationships between currents, temperature, and calibrated ocean color products from MASS illuminated the influence of submesoscale dynamics on phytoplankton ecosystems. This study highlights the potential of airborne remote sensing to quantify the submesoscale processes that structure phytoplankton ecosystems and oceanic carbon, without the limitations impacting in-water sensors.

1. Introduction

Satellites and airborne sensors reveal rich submesoscale (1–10 km) variability in ocean color marked by filamentous structures, eddies, and patches arising from variations in phytoplankton concentration and pigments at the sea surface. Multi- and hyperspectral ocean color provides proxies for chlorophyll-a (chl-a, indicating phytoplankton abundance) and other pigments that may reflect changes in community structure (Chase et al., 2017). The variability in ocean color is closely tied to physical dynamics that restructure phytoplankton ecosystems and drive active biological responses such as changes in primary productivity and shifts in community structure (Lévy et al., 2018, 2024; Mahadevan, 2016; Pereira et al., 2024). As the base of the marine food web and a key component of the biological carbon pump, phytoplankton are crucial to the overall health of marine ecosystems and to the ocean's role in climate (Siegel et al., 2023). New tools to simultaneously document small-scale phytoplankton distributions and corresponding physical dynamics are critical for disentangling the role of submesoscale processes in ocean biogeochemistry and climate.

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Submesoscale processes are defined by a Rossby number (Ro) of O(1), length scales of about 1–10 km, and time scales on the order of hours and days (McWilliams, 2016; Thomas et al., 2008). Ageostrophic dynamics at Ro of O(1) can give rise to significant vertical velocities (Mahadevan & Tandon, 2006; Thomas et al., 2008), which induce fluxes of oxygen, nutrients, and carbon that lead to biogeochemical impacts (Omand et al., 2015; Zhang et al., 2019). Submesoscale dynamics are spatially heterogeneous, quickly evolving, and occur on similar timescales to phytoplankton growth and mortality. This presents a difficult observational challenge to characterizing the relationships between dynamics and phytoplankton distributions. The slow speed of many in situ sampling platforms (e.g., ships and autonomous vehicles) relative to time scales of variability can lead to spatiotemporal aliasing (Lenain et al., 2023). Satellite measurements can augment in situ observations, but have too coarse spatial resolutions to resolve 1–10 km features (e.g., NASA SWOT achieves a spatial resolution of O (1 km); Archer et al., 2025). Airborne remote sensing capturing simultaneous measurements of ocean color and ocean physics provides a powerful tool to capture submesoscale distributions and dynamics by measuring high resolution snapshots.

The Modular Aerial Sensing System (MASS; Melville et al., 2016), from the Air-Sea Interaction Laboratory at the Scripps Institution of Oceanography, is a portable airborne package of instruments measuring and estimating concurrent, high resolution (256 m spatial resolution) ocean currents (Freilich et al., 2023; Lenain et al., 2023), surface waves (Lenain & Melville, 2017; Lenain & Pizzo, 2020), sea surface slope statistics (Lenain et al., 2019), internal waves (Lenain & Pizzo, 2021), sea surface height (Villas Bôas et al., 2022), sea surface temperature (SST; Lenain & Pizzo, 2021), Stokes drift (Lenain & Pizzo, 2020), and hyperspectral ocean color at a 3.4 nm spectral resolution. Measurements of interest in this study are surface currents, SST, and hyperspectral ocean color from MASS.

Concurrent bio-optical and physical measurements from the same airborne platform capture the relationships between fine-scale biology and physics with no temporal separation. Vorticity, divergence, strain, and kinetic energy flux can be calculated from synoptic velocity measurements to characterize submesoscale dynamics (Freilich et al., 2023; Lenain et al., 2023), identify flow regimes (Balwada et al., 2021), and quantify energy cascades that can strengthen the mesoscale or transfer energy to dissipation scales (Balwada et al., 2022; D'Asaro et al., 2011; Freilich et al., 2023). For example, submesoscale frontogenesis is associated with strong surface convergence (negative divergence), enhanced positive vorticity, high strain, and enhanced kinetic energy flux to dissipation scales (Barkan et al., 2019; D'Asaro et al., 2011; Srinivasan et al., 2023). This surface signature is associated with an ageostrophic secondary circulation in the vertical, which induces upwelling on the less dense side of the front and downwelling on the dense side (Mahadevan, 2016). Eddies are associated with high absolute values of vorticity and weaker strain (Balwada et al., 2021). Relationships between phytoplankton and divergence may arise due to the impacts of vertical velocities on phytoplankton growth and community structure (Plummer & Freilich et al., 2023) or between vorticity and phytoplankton due to coherent trapping by eddies (Jones-Kellett & Follows, 2024).

Here, we present the first airborne observations pairing concurrent snapshots of sub-kilometer ocean velocities and their derivatives (i.e., vorticity, divergence, and strain) with ocean color. We developed proxies for chl-a and particulate organic carbon (POC) using the hyperspectral ocean color data from MASS, statistical modeling, and in situ evaluation without using atmospheric correction. The development of ocean color methods for MASS provides an exciting opportunity to merge state-of-the-art measurements of submesoscale dynamics (Lenain et al., 2023) with high-resolution ocean color data to better understanding the submesoscale processes underpinning biological distributions and biogeochemical processes. We also explore five examples of submesoscale features, with signatures in ocean color, SST, and velocity derivatives, to highlight potential impacts of submesoscale dynamics on phytoplankton.

2. Data and Methodology

2.1. Experiment Overview and Study Area

The data in this study were collected during the pilot campaign of the Sub-Mesoscale Ocean Dynamics Experiment (S-MODE), a NASA Earth Venture Suborbital-3 (EVS-3) mission. The overall goal of S-MODE is to test the hypothesis that submesoscale ocean dynamics drive significant vertical exchanges in the upper ocean (Farrar et al., 2020). Measurements from various airborne sensors are complemented by research vessels, autonomous vehicles (e.g., Wavegliders, Saildrones, subsurface neutrally buoyant Lagrangian floats, surface drifters, gliders), satellite data, and high resolution ocean models to diagnose and quantify the submesoscale dynamics responsible for vertical transport.

The S-MODE study site extends 200 km offshore of San Francisco in the central California Current System (CCS, Figure 1), a region known for strong seasonal upwelling. Upwelling of cold, nutrient-rich water near the coast drives high levels of biological productivity (García-Reyes & Largier, 2012), and sets large gradients in properties such as temperature, salinity, and chlorophyll. Westward-propagating mesoscale eddies (10–100 km) transport water away from the coast (Amos et al., 2019) and support the formation of submesoscale features, which are

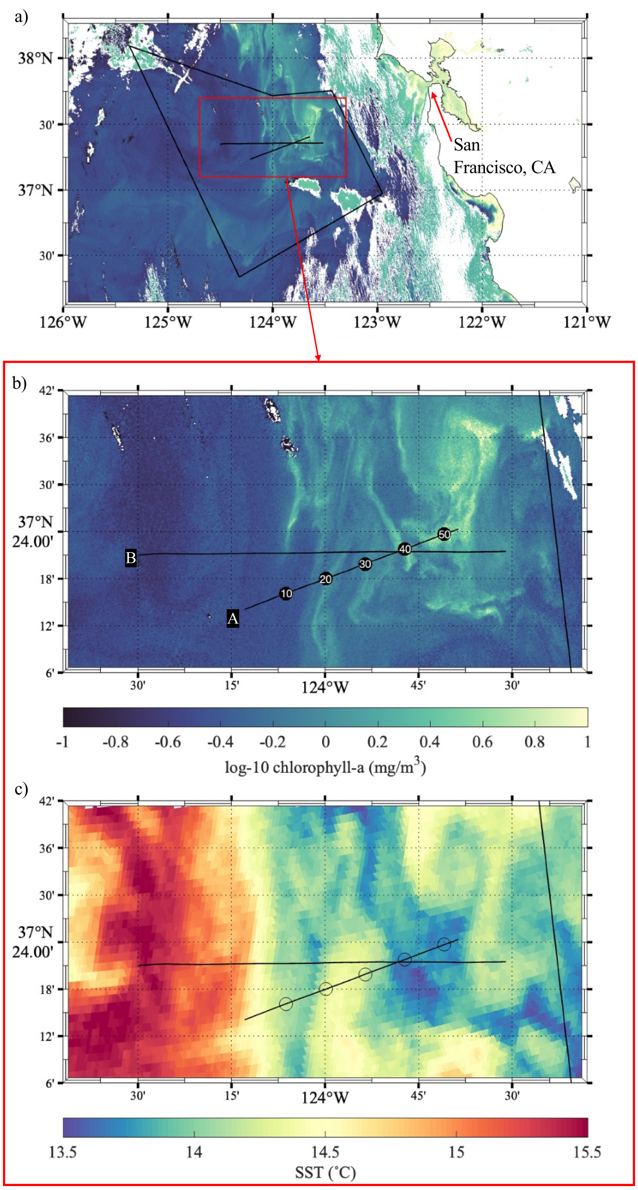

Figure 1. (a) The S-MODE study region (black polygon) is overlaid on Sentinel-3 log-10 chlorophyll-a from 29 October 2021 (0.3 km resolution, time of collection (TOC): 18:44 – 18:47 UTC). Black lines indicate the repeated vessel and MASS transects. (b) A zoomed-in view of Transects A and B overlaid on Sentinel-3 chl-a and (b) sea surface temperature (c) from the same day (1 km resolution, TOC: 17:01 – 18:42Z). Transect A was flown over 5 times and the lines will be referred to as A1 (TOC: 19:36 – 19:57 UTC), A2 (TOC: 20:23 – 20:41 UTC), A3 (TOC: 20:44 – 21:01 UTC), A4 (TOC: 21:06 – 21:23 UTC), and A5 (TOC: 21:52 – 22:09 UTC). Transect B was flown over 2 times and the lines will be referred to as B1 (TOC: 20:31 – 20:51 UTC) and B2 (TOC: 21:01 – 21:28 UTC). Flights over these transects made parallel overlaps with the ship (TOC: 12:44 – 21:29 UTC). Distance along track in kilometers for Transect A (line A2 analyzed in Sections 5 and 6) are denoted by black circles and white numbers in panel (b).

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ubiquitous in the CCS (Checkley & Bath, 2009). Sharp gradients in phytoplankton species have been observed across submesoscale fronts in the CCS (Mangolte et al., 2023; Taylor et al., 2012). However, relationships between fronts and phytoplankton diversity are variable and depend on the underlying mechanisms and resulting ecological responses (Lévy et al., 2015, 2018).

The S-MODE pilot campaign took place between 22 October – 8 November 2021, during which MASS collected hyperspectral data over 32.55 hr and 5,997.2 km. There were 7 parallel overlaps with the ship that were within 2 hr and during suitable times for ocean color data collection. Overlaps occurred directly over the ship, allowing co-located match-ups between ship-data and airborne data to be co-located. Five of these overlaps occurred on 29 October (Transect A; lines A1, A2, A3, A4, A5; Figure 1) and two occurred on 30 October (Transect B; lines B1, B2; Figure 1). Parallel overlaps with the ship allowed for the development of empirical chl-a and POC proxies from in situ observations. The repetition of transects increases confidence in the variability observed and provides an opportunity to observe the evolution of fine-scale features.

2.2. Data Collection and Processing

2.2.1. In Situ Water Samples and Bio-Optical Proxies

Seawater samples for chl-a and POC ($N = 149$) were collected from the R/V Oceanus science seawater system at predetermined intervals during the S-MODE pilot cruise and analyzed with a benchtop Turner A10 fluorometer (chl-a) and a Costech 4010 Elemental Analyzer (POC). Seawater was pumped through a diaphragm pump to prevent damage to phytoplankton cells. Seawater samples were used to create in situ proxies of chl-a and POC from flow-through bio-optical data. Flow-through bio-optical data were collected from a WETStar Fluorometer (Seabird Scientific) measuring chlorophyll fluorescence (excitation at 460 nm, emission at 695 nm) and a C-Star Transmissometer (Seabird Scientific) measuring beam transmittance (converted to beam attenuation, c_p) with a 25 cm pathlength. To develop a chl-a optical proxy, we first scaled c_p to nighttime chl-fl (isolated with a photosynthetically active radiation threshold) using a logarithmic relationship ($R^2 = 66.6\%$). Scaled c_p had a significant linear relationship with bottle measurements of chl-a ($R^2 = 72\%$). c_p and nighttime chl-fl were used to create the bio-optical chl-a proxy to avoid the effects of non-photochemical quenching (NPQ). NPQ is a phytoplankton physiological response to excess light conditions, which causes decreased fluorescence given the same amount of chlorophyll (Xing et al., 2018). A power law fit between c_p and POC was used as a bio-optical proxy for POC (Bishop, 1999; $R^2 = 72.8\%$). Detailed descriptions of in situ methods are described in Lang et al. (2023).

2.2.2. Airborne Hyperspectral Ocean Color Data

Airborne data considered in this study were collected at two discrete altitudes: 0.4 km ("low altitude") and 0.9 km ("high altitude"). MASS can fly underneath high altitude clouds. As long as there is still sufficient solar irradiance for ocean color retrievals, MASS can capture ocean color data at times that orbital sensors and high-altitude airborne sensors are unable. MASS is equipped with a downward looking AISA Kestrel 10 Camera, measuring upwelling radiance ($L_t(\lambda)$; λ = wavelength of interest in nm). The AISA Kestrel 10 is a pushbroom imager with a spectral range of 398.93–1,004.62 nm at a 3.4 nm spectral resolution and a 1,024 pixel cross-track swath. Native spatial resolutions are sub-meter-scale at flight altitudes of 0.4–1 km. Polarization sensitivity is $<\pm 2\%$ and the field of view is 40°. An upward looking Fiber Optic Downwelling Irradiance Sensor (FODIS) (SPECIM, SPECTRAL IMAGING LTD.) measuring diffuse downwelling irradiance ($E_d(\lambda)$) at a ~0.6 nm spectral resolution was mounted on the top of the aircraft.

Raw $L_t(\lambda)$ pixels were spatially averaged with a 15 × 32 box (corresponding to about 9 × 13 m resolution at 0.4 km altitude and 10 × 22 m resolution at 0.9 km altitude). $E_d(\lambda)$ data were smoothed with a 2 min median filter. Cosine corrections of Homolova et al. (2009) were applied to Ed data to account for varying viewing zenith and azimuth angles introduced by the motion and tilt of the aircraft.

Ocean color remote sensing data typically undergo an atmospheric correction. Atmospheric correction isolates the water-leaving radiance from the total upwelling radiance signal $L_t(\lambda)$ by removing any contributions from the atmosphere (above and below the sensor) and sea surface (e.g., glint, white-caps, background sky radiance reflected off the sea surface) (Mobley et al., 2016). This procedure also accounts for varying solar and viewing geometries, out-of-band responses, and polarization effects (Mobley et al., 2016). Atmospheric correction is a crucial step for satellite ocean color retrievals, as the atmosphere contributes ~70%–90% of the total top of the atmosphere signal (Mobley et al., 2016). For low-altitude sensors, atmospheric scattering below the sensor becomes less significant.

Airborne atmospheric correction is different than satellite atmospheric correction in its procedures and challenges. e.g., ancillary atmospheric data is sometimes used to augment the estimation of the atmospheric contribution to satellite retrievals (e.g., NASA l2gen, POLYMER; Baith et al., 2001; Steinmetz et al., 2011). These data are too coarse to be used for airborne ocean color retrievals in the same way. Additionally, Cox-Munk statistics are used to estimate the glint contribution from wind waves (e.g., NASA l2gen, POLYMER; Baith et al., 2001; Steinmetz et al., 2011). These models are not applicable at the small spatial scales captured by MASS because they depend on relationships defined at large scales (>100 m) and do not account for glint due to surface roughness (Wang et al., 2024). Lastly, (sub) meter resolutions are computationally expensive because atmospheric correction is performed on individual pixels. To avoid introducing errors that would be difficult to track through a complicated and computationally expensive atmospheric correction, we evaluated if ocean color measurements from MASS could capture significant variability in chl-a and POC, without atmospheric correction. Because MASS is flying at low altitudes (0.4–0.9 km altitudes for MASS vs. >600 km for ocean color satellites), the atmosphere has a much smaller contribution to the total signal. Instead, we implemented cloud masking and manually removed high glint, used statistical methods to account for changing solar and viewing geometries, used direct measurements of $E_d(\lambda)$ to partially account for changing atmospheric conditions above the aircraft, and evaluated ocean color estimations with in situ bio-optical data.

A cloud mask was generated with RGB bands and support vector machines (SVM) following the methods of Kang et al. (2019). Cloud masks were visually inspected and compared to RGB images generated with the red (658 nm), green (555 nm), and blue (452 nm) bands of the hyperspectral data. Significant cloud coverage was detected in the first and last ~10 km of the 29 October flight lines (A1–A5) by the SVM masks, and further verified by RGB images and flight logs. These regions were masked for all analyses. This comparison also indicated that the mask falsely identified some regions of high sun glint as clouds in the middle of tracks. In these regions, ocean color variability was still linked to ship-based chl-a and POC variability, but likely impacted measurements by raising the magnitude of retrievals. We applied the cloud mask to all data used in modeling airborne chl-a and POC proxies (Section 4) to subset the highest quality data, but ignored the cloud mask for lines A1–A3 (clear atmospheric conditions) for following analyses to avoid data gaps.

Very high sun glint contamination occurred on the edges of the swaths and were manually removed. This type of sun glint can be visually detected in visible bands as high radiance values that run parallel to the flight path and along the edges of the swath. This corresponded to an average of ~200 m (~25% of the ocean color swath) along the top of high altitude flights and 80 m (~20% of the ocean color swath) along the top of low altitude flights.

Surface reflectances ($r_{rs}(\lambda)$, sr^{-1}) were found by taking the ratio of $L_t(\lambda)$ to $E_d(\lambda)$. Data were further quality-controlled to only include data with no low-altitude clouds, during times with optimal solar zenith angles to minimize glint (<60°), and under conditions that did not include white caps. Atmospheric conditions and wave conditions were qualitatively logged during data collection. Flight logs were used to further ensure that data from flights with sub-optimal conditions were not included in the analysis. $L_t(\lambda)$, $E_d(\lambda)$, and $r_{rs}(\lambda)$ spectra from line A2 plotted in Figure S1 of Supporting Information S1.

2.2.3. Airborne Sea Surface Temperature and Surface Currents

Sea surface temperature (SST) maps were produced using a Flir SC6700SLS longwave infrared (LWIR) camera at a 1 m resolution (Freilich et al., 2023; Lenain & Pizzo, 2021). Surface currents were obtained with the "DoppVis" instrument, which uses a Nikon D850 camera synchronized with a coupled Global Positioning System/Inertial Motion Unit (GPS/IMU) system to capture the spatiotemporal evolution of surface waves in visible imagery. Lagrangian mean current profiles are inferred from the difference between the expected wave frequency in the absence of currents and the observed wave frequency, modulated by the underlying current (Lenain et al., 2023). Lines A2 and A3 were high-altitude flights, which is necessary for the collection of cross-swath velocity data and the calculation of 2D spatial gradients to yield vorticity, divergence, and strain. A2 had the

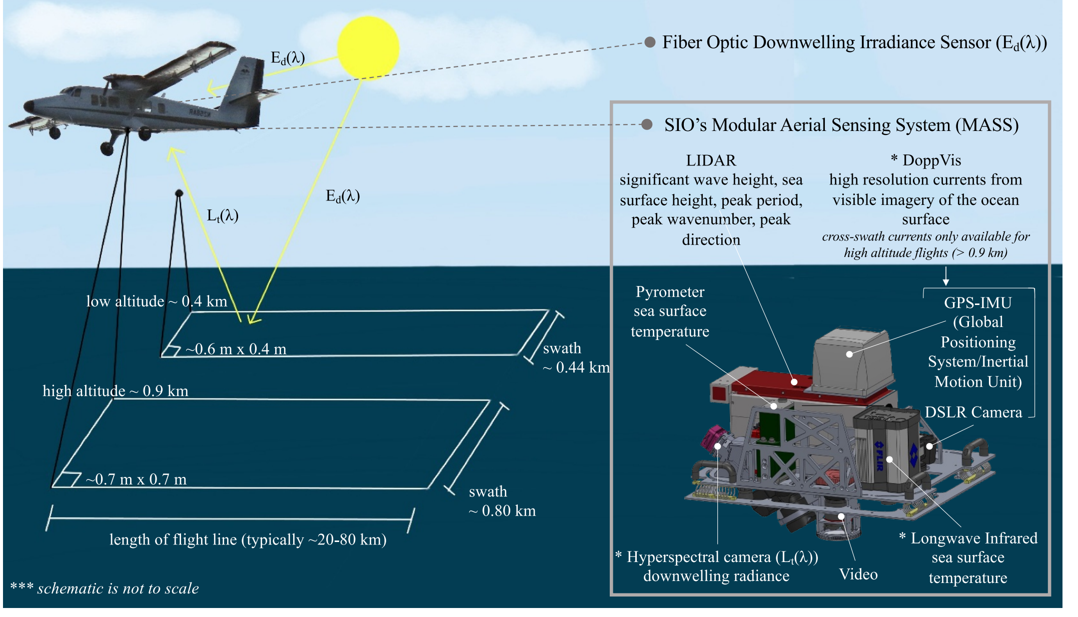

Figure 2. Schematic describing the dependency of the swath width and spatial resolution of ocean color data on the altitude of the Twin Otter aircraft (top left). The hyperspectral camera on MASS faces downwards, measuring upwelling radiance ($L_t(\lambda)$) from the sea surface and the atmosphere below. FODIS measures hyperspectral downwelling irradiance ($E_d(\lambda)$) and is mounted on the top of the aircraft. MASS is equipped with a lidar, DoppVis instrument, pyrometer, hyperspectral camera, video, and longwave infrared (right box). MASS instruments analyzed in this study are starred. Schematic is not to scale.

clearer atmospheric conditions of the high altitude flights and is the focus of analysis and discussion in Sections 3.3 and 3.4.

Currents from A2 were binned to a 256 m × 256 m resolution (Freilich et al., 2023). Cross-swath zonal and meridional velocity components (u, v) from DoppVis were converted to along-track (x) and across-track (y) velocity components (u', v'), and the coordinate system was rotated into the along-track direction. Vorticity ($\zeta = v'_x - u'_y$), divergence ($\delta = u'_x + v'_y$), and strain ($\sigma = [(u'_x - v'_y)^2) + (v'_x + u'_y)^2)]^{1/2}$) were calculated from u' and v'. Vorticity, divergence, and strain were normalized by the Coriolis parameter ($f = 0.89 \times 10^{-4}\ \mathrm{s}^{-1}$).

Kinetic energy (KE) flux was also calculated from DoppVis data as in Freilich et al. (2023). Instantaneous KE flux was calculated using a coarse-graining approach (Aluie et al., 2018; Eyink, 2005; Germano, 1992):

$$\pi = -(\tau_{u'v'}(\bar{u}'_y + \bar{v}'_x) + \tau_{u'u'}\bar{u}'_x + \tau_{v'v'}\bar{v}'_y \tag{1}$$

where $\tau_{ab} = \overline{ab} - \bar{a}\bar{b}$ and $\bar{\cdot}$ is a top hat filter with a 1 km scale as in Freilich et al. (2023). Positive KE flux indicates a forward cascade toward smaller scales and negative KE flux indicates an inverse cascade toward larger scales.

2.3. Modeling Airborne chl-a and POC Proxies

Regional airborne chl-a and POC algorithms were empirically derived using in situ optical-proxies. Hyperspectral data was first matched-up with corresponding ship-based proxies of chl-a and POC at the same location and within 2 hr, yielding 700 match-ups. One-minute binned flow-through bio-optical proxies of log-10 chl-a and log-10 POC were used as the dependent variables to account for their log normal distributions (Craig et al., 2012). "Band ratios" were the primary predictors of the bio-optical proxies and based on algorithms from NASA's Ocean Biology Processing Group (OBPG): $r_{rs}(486)/r_{rs}(555)$ for chl-a and $r_{rs}(443)/r_{rs}(555)$ for POC (O'Reilly et al., 1998; O'Reilly & Werdell, 2019; Stramski et al., 2008). Band ratios from the center of the swath were selected for match-up with in situ proxies.

We detected significant spatial autocorrelation between log-10 chl-a and log-10 POC and band ratios using autocorrelation functions. To reduce autocorrelation, we sub-sampled the data at the shortest interval (8 points) which reduced autocorrelation to be non-significant after the first lag. Sub-sampling yielded 89 match-ups for multiple regression (gray highlighted points in Figure 3). Chl-a and POC were estimated with multiple regression models using a "band ratio" and ancillary airborne variables as inputs.

To account for changing viewing and solar geometries between MASS overpasses, the following ancillary predictors were explored: plane heading, plane altitude, solar azimuth angle (SAA), and solar zenith angle (SZA). Plane heading and altitude were smoothed using a 10-min moving mean filter to avoid introducing noise into model estimates. The largest changes in these parameters occurred between flights (durations ≥10 min). All predictor variables were centered by subtracting the mean. The variance inflation factor (VIF) was calculated for a full set of potential predictors to test for multicollinearity. Solar azimuth angle was dropped due to its high VIF. VIFs were reduced to low values (<4). Backwards model selection and type III ordinary least squares linear multiple regression was used to predict log-10 chl-a from band ratios, plane heading, plane altitude, SZA, and their interactions. Non-significant interaction terms ($P < 0.05$) were removed. The procedure was repeated for log-10 POC. We also investigated the use of higher order fits for chl-a and a power law relationship for POC, following NASA's OCx chl-a and POC algorithms (O'Reilly et al., 1998; O'Reilly & Werdell, 2019; Stramski et al., 2008).

2.4. Principal Component Analysis on Hyperspectral Data

We investigated the use of the full hyperspectral data to illuminate shifts in phytoplankton community groups. Hyperspectral ocean color sensors capture continuous spectral information, allowing for the estimation of phytoplankton pigment groups from unique spectral signatures given by combinations of photosynthetic accessory pigments (Kramer et al., 2022). Sufficient in situ accessory pigment data was not available during the pilot experiment for the validation of phytoplankton pigment-based algorithms beyond chl-a. Instead, principal component analysis (PCA) was used to decompose the spectra into signatures from different optical constituents which could include phytoplankton accessory pigments. PC's were interpreted based on their spectral shapes. The inputs to the PCA were the nadir values of $r_{rs}(\lambda)$ between 400 and 720 nm from A1–A5. The spectrum above 720 nm becomes more heavily affected by atmospheric constituents such as absorption by water vapor or oxygen (Mobley et al., 2016). Each $r_{rs}(\lambda)$ was standardized by first removing the mean spectra. Each $r_{rs}(\lambda)$ was then normalized by the integral of $r_{rs}(\lambda)$ between 400 and 720 nm as in Craig et al. (2012) and the mean of the normalized spectra was subtracted.

3. Results and Discussion

3.1. Airborne chl-a and POC Models

Chl-a and POC multiple regression models were significant (chl-a: $F_{5,83} = 40.6$, $P < 0.001$, $R^2 = 66.2\%$; POC: $F_{3,85} = 54.5$, $P < 0.001$, $R^2 = 56.2\%$). Root mean square errors (RMSE) were low for both models (chl-a: 0.12; POC: 0.11). Model parameters and statistics associated with the multiple regression models for chl-a and POC are given in Tables S1 and S2 of Supporting Information S1. Model assumptions were met. A comparison of normalized residuals against model predictions showed homogeneity of variance (not shown). Histograms and qq plots showed normality of residuals (not shown).

Band ratios were significantly correlated with log-10 chl-a and log-10 POC ($R^2 = 56.8\%$, $R^2 = 46.2\%$; Figures 3a and 3b), but the inclusion of ancillary parameters in the models yielded higher R^2 values (chl-a: $R^2 = 66.2\%$; POC: $R^2 = 56.2\%$) and lower Akaike information criterions (AIC, lowest value indicates best fit; chl-a: −123.1; POC: −132.3) than simple linear fits between band ratios and log-10 chl-a and POC (chl-a AIC: −107.6; POC AIC: −116.3). Linear models were also more appropriate than higher order fits for log-10 chl-a and a power law relationship for log-10 POC (log-10 chl-a: 1st order AIC = −107.6; 2nd order AIC = −105.7; 3rd order AIC = −106.4; 4th order AIC = −104.9; log-10 POC: power law: AIC = 605.6; linear fit: AIC = −132.3).

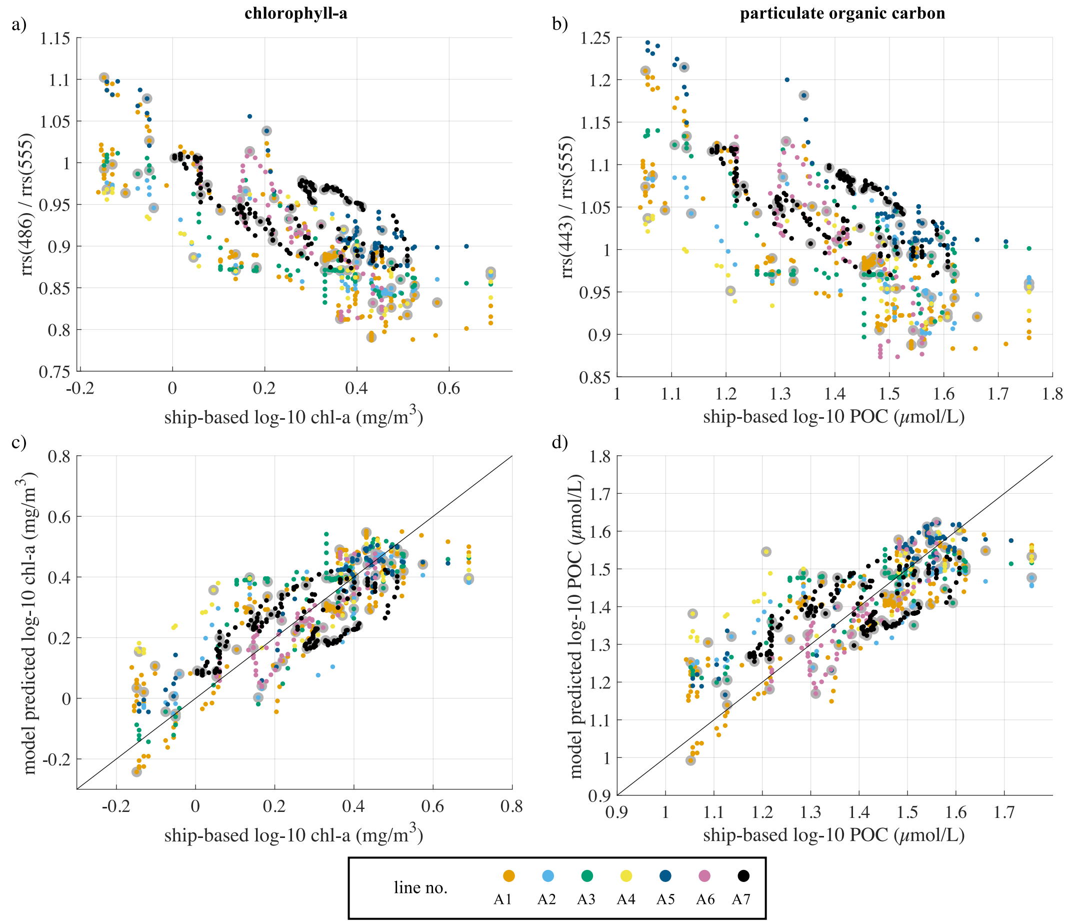

Figure 3. Band ratios are plotted against corresponding in situ measurements, log-10 chlorophyll-a (a) and log-10 particulate organic carbon (b). Airborne model predicted log-10 chlorophyll-a plotted against corresponding ship-based log-10 chlorophyll-a (c). Airborne model predicted log-10 particulate organic carbon plotted against corresponding ship-based log-10 particulate organic carbon (d). Points are colored by line number. Data points used to train the multiple regression model are outlined in light gray. 1:1 lines in solid black.

Chl-a and POC models were applied to full airborne swaths. Variability in along-track chl-a and POC at $y = 300$ m for the three clearest and longest repeat lines (A1–A3; Figures 4a and 4b) agree well with ship-based chl-a and POC (Figure 4c). Variability also agrees well with Sentinel-3 OLCI chlorophyll-a (0.3 km resolution), albeit measuring different scales ($R^2 = 63.7\%$). However, chl-a and POC models predict larger magnitudes than ship-based measurements and Sentinel-3 OLCI.

Cross-swath chl-a, captured by repeat overpasses, can capture sub-kilometer chl-a features such as the feature shown in Figures 4d–4f. Successive flights measure a snapshot of the same feature three times in ~1 hr. Overall, the spatial variability captured by the overpasses agree well. Some differences arise due to their differing N-S extents and temporal separation. For example, A1 has a narrower swath due to its higher altitude (~200 m swath vs. ~400 m swath), and A3 is shifted about 50–100 m North relative to A2. Between 38 and 39.5 km, a thin filament of elevated chl-a separates two patches of low chl-a water. The feature also seems to have experienced a temporal evolution between overpasses and potentially an advection down-swath. Additionally, each swath captures a high chl-a region between 39.4 and 41.4 km surrounded by low chl-a water. There is a peak in chl-a between 40.5 and 41 km, and the dynamical signatures of this feature are further explored in Section 7.1.3. The ship also detects these chl-a features, but only MASS can observe the 2-D structures of the patches.

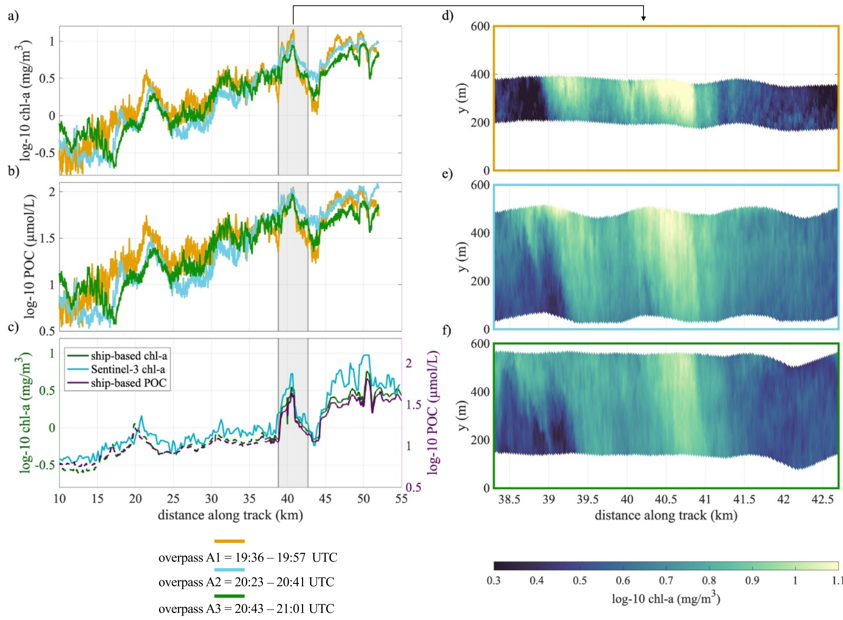

Figure 4. Along-track log-10 chl-a (a) and log-10 POC (b) for the three longest and clearest lines: A1 (orange), A2 (blue), and A3 (green). One-second binned ship-based chl-a (dark green) and POC (purple) (c). Ship based observations greater than 2 hr apart from plane observations denoted by dashed lines in panel (c), and ship observations within 2 hr denoted by solid lines in panel (c). Full swath view of high-chlorophyll feature (d–f) with central line time in UTC labeled below swath.

3.2. Spectral Signatures From Hyperspectral Data

Principal component 1 (PC1) explained 88.0% of the covariance in the spectra and may be attributed to particulate backscattering based on its spectral shape (Lange et al., 2020; Figure 5a). PC2 explained 8.51% and may be associated with chl-a as in Lange et al. (2020) due to peaks at ∼561 nm where chlorophyll-a reflects light and at ∼682 nm where chl-a has its fluorescence emission peak (Craig et al., 2012; Figure 5b). PC3, PC4, and PC5 explained 2.71%, 0.39%, and 0.07%. Although accounting for a small part of the variability, PCs 3–5 contain peaks at ∼680–690 nm and/or ∼560–570 nm possibly associated with fluorescence of phytoplankton (Sathyendranath et al., 1994; Vishnu et al., 2022) and variability in phytoplankton accessory pigments. PC3 contains peaks at ∼568 nm and ∼685 nm (similar to PC2), PC4 contains peaks at ∼500 nm and ∼692 nm, and PC5 contains peaks at ∼568 nm (similar to PC3) and ∼692 nm. PC6 and higher may be attributed to less abundant accessory pigments or atmospheric parameters (PC spectrum shown in Figure S2 of Supporting Information S1). The signs of PC2 and PC5 were flipped for interpretability, because the directions of PCA loadings are arbitrary. We determined their direction of change based on fluorescence peaks around ∼560–570 nm and ∼680 nm.

The relationships between PC1-5 and log-10 MASS chl-a and log-10 ship chl-a were explored for line A2 to further test if PC's were capturing the signature of in-water optical constituents. PC2 was strongly correlated with log-10 MASS chl-a (Figure 5e, $R^2 = 77.7\%$) and log-10 ship chl-a (Figure 5g, $R^2 = 51.0\%$). Other lines had strong relationships between ship-based chl-a and PC2 (Figure S3 in Supporting Information S1), but there was more variability between lines when using PC2 than when using the band-ratio. The relationship between MASS chl-a and PC2 was strengthened by adding PC1 and PCs 3–5 as predictors ($R^2 = 98.2\%$) and between ship chl-a and PC2 ($R^2 = 70.2\%$). Waters with high PC1 values (orange and red points in 5f) have different slopes than low PC1 values (yellow, green, and blue points in 5f) between PC2 and log-10 MASS chl-a. PC1 and PC3-5 strengthened the relationships between PC2 and chl-a estimated by MASS and the ship because both chl-a proxies (band ratio for MASS, bio-optics for the ship) were likely affected by optically active constituents of the water column represented by PC1 and PC3-5. In Sections 3.3 and 3.4, PC3-5 will be used to infer shifts in phytoplankton accessory pigments, and hence, different phytoplankton community types.

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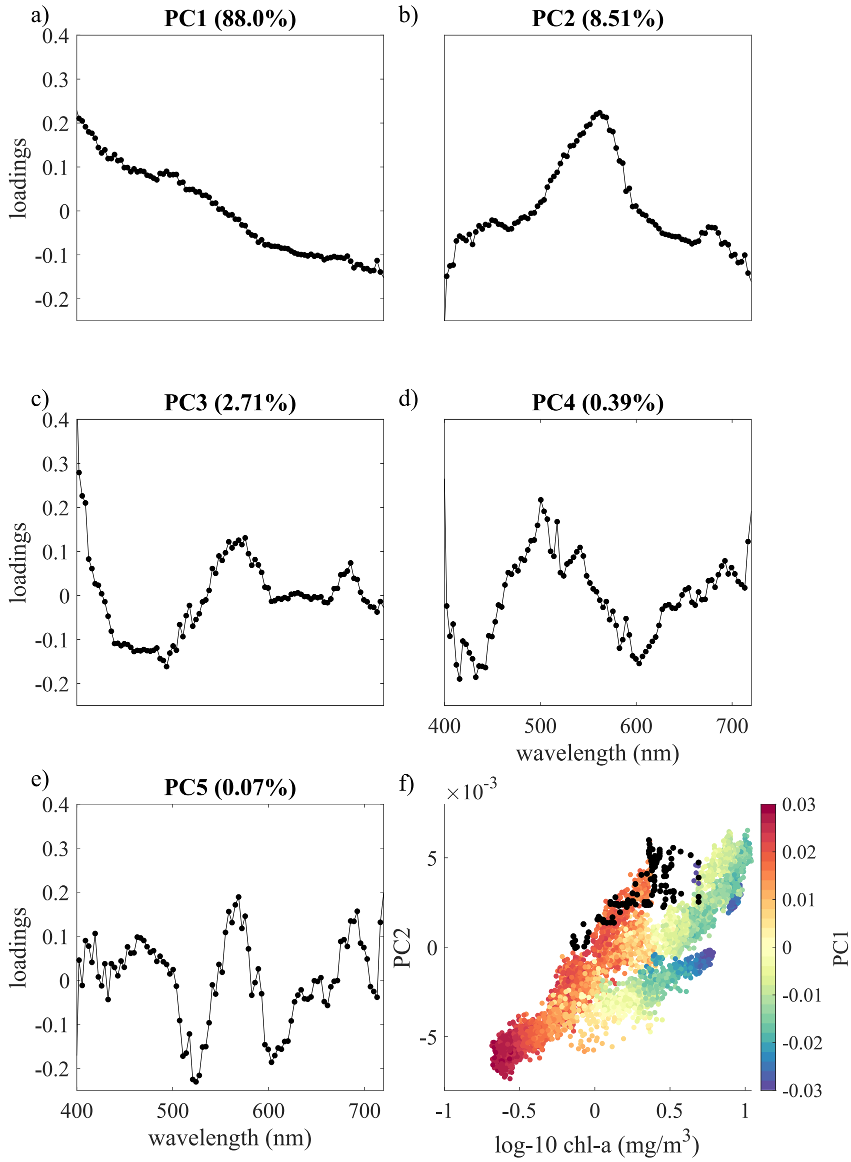

Figure 5. Loadings and percent variances for PC1, PC2, PC3, PC4, and PC5 (a–e). PC2 plotted against log-10 MASS chlorophyll-a (colored points, f) and ship-based log-10 chlorophyll-a (black points, f) for line A2.

3.3. Direct Observations of Sea Surface Temperature, Ocean Color, and Current Derivatives at Submesoscale Features

Concurrent measurements of ocean color, SST, and currents provide the opportunity to uncover relationships between submesoscale phytoplankton variability, SST, and surface currents. First, in situ temperature, salinity, and density along transect A was used to examine the surface density structure underlying the temperature gradients measured by MASS. Typically, temperature fronts are strong indicators of density fronts in this region due to the ubiquity of cold, dense, upwelling fronts in the California Current System (Mauzole et al., 2020). However during the campaign, density was dominated by salinity (Figure 6a) due to freshwater input (e.g., precipitation, river outflow) following an atmospheric river event that occurred 24–25 October. Increased chl-a was associated with both positive (F1, F2, F3, F5 in Figures 6b–6d) and negative SST gradients (F4 in Figures 6b–6d), indicating changes in the hydrographic and biological properties of typically colder, saltier, and higher chl-a waters sourced from coastal upwelling and surrounding waters typically warmer, fresher, and lower in chl-a.

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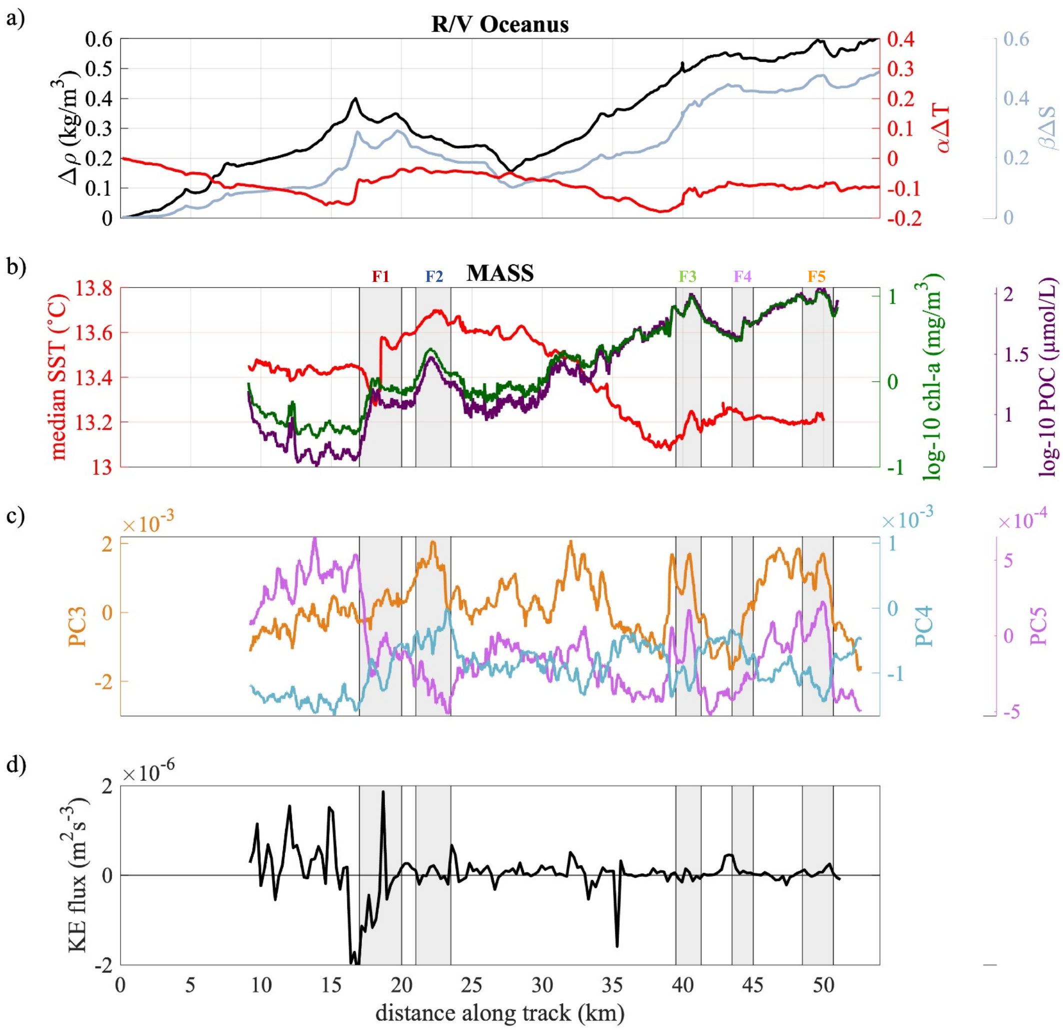

Figure 6. $\Delta\rho$ (black), $\beta\Delta S$ (gray), $\alpha\Delta T$ (red) measured by the R/V Oceanus plotted against distance along the track (a). $\Delta\rho$ denotes the density change from the start of the track, where distance along track is 0 km. ΔT and ΔS are the changes in temperature and salinity from the start of the track. ΔT and ΔS are scaled by their corresponding thermal expansion and salinity contraction coefficients (α, β). SST (red), log-10 chlorophyll-a (dark green), and log-10 particulate organic carbon (purple) plotted against distance along track (b). PC3, PC4, and PC5, representing spectral shifts potentially associated with phytoplankton community shifts, plotted against distance along track (c). Kinetic energy flux plotted against distance along track (d). Five features of interest plotted in Figures 8–11 denoted by gray shaded regions (b–d).

The ship took about 9.5 hr to complete transect A (track shown in Figure 1b, data plotted in Figure 6a) and MASS took 20 min to complete line A2 (Figures 6b–6d). Ship-based observations captured spatial and diurnal variability, whereas MASS-based observations captured the spatial variability of a snapshot in time. Fine-scale features evolved and moved quicker than the ship completed the transect. For example, the temperature front (F1 in Figures 6b–6d and ~17 km along track in Figure 6a) shifted ~2.0 km along the track in the ~5.4 hr between the ship's crossing and MASS flight. Airborne measurements avoid spatiotemporal aliasing that affects in situ platforms. MASS also measures the skin temperature whereas ship-based temperature is measured at several meters depth.

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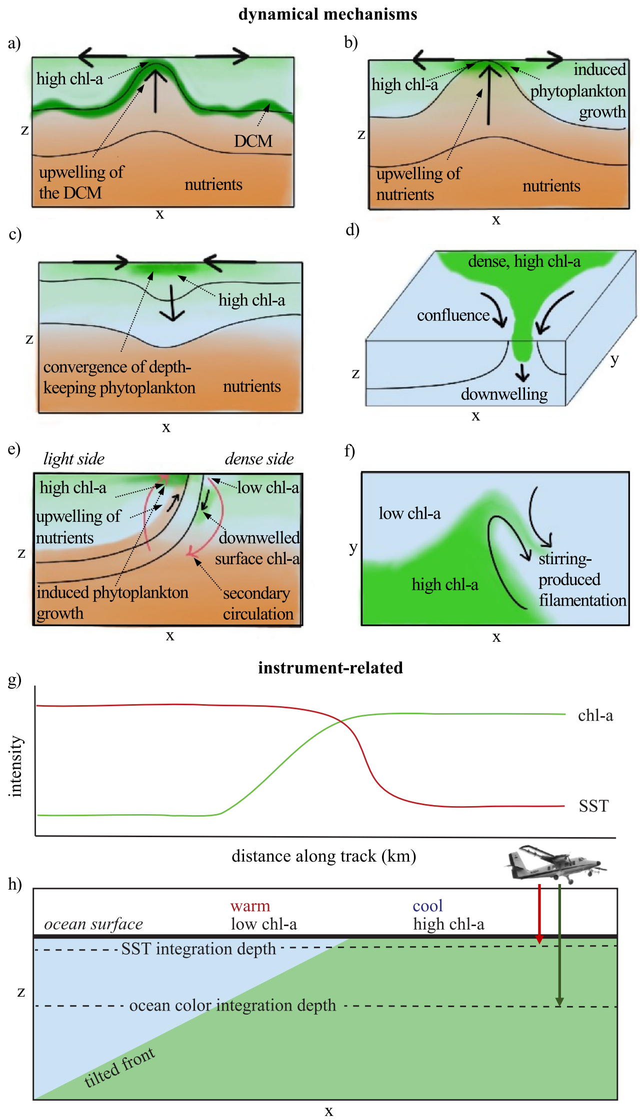

Figure 7. Schematics that illustrate a few of the dynamical mechanisms driving local surface chl-a enhancement or reduction (a–f). Upwelling of the deep chlorophyll maximum (DCM) induced by surface divergence brings high chl-a water toward the surface (a). Upwelling of nutrients induced by surface divergence can spur the growth of phytoplankton in nutrient-limited regimes (b). Surface convergence of phytoplankton at the surface can lead to high chl-a patches in regions of downwelling (c). Dense filaments pinching off of high chl-a water masses can converge and downwell, leading to enhanced surface chl-a associated with a vertical extent (d). Ageostrophic secondary circulation (red arrows) associated with submesoscale fronts can induce vertical velocities impacting surface phytoplankton concentrations (e). Stirring of larger chl-a gradients can generate submesoscale filamentous structure (f). Schematics that demonstrate how different measurement integration depths between sensors can affect ocean color and SST measurements from MASS (g, h). Approaching the front from the warm side, the chl-a front is detected by MASS before the SST front due to its deeper integration depth. Integration depths can also cause SST gradients to be sharper than chl-a gradients.

3.4. Dynamical Influences on the Observed Biophysical Distributions

Numerous dynamical processes can lead to surface phytoplankton variability and co-variations with sea surface temperature and current derivatives. For example, upward vertical velocities, associated with surface divergence, can upwell nutrients from the pycnocline into the surface mixed layer and result in enhanced phytoplankton production at the surface in nutrient-limited regimes (Mahadevan, 2016; Figure 7a). Upwelling can also shift a deep chlorophyll maximum (DCM) from the nutricline toward the surface, increasing their exposure to light and increasing surface concentrations (Lévy et al., 2018) if the DCM is a biomass maxima as well (as opposed to a chlorophyll-fluorescence maxima due to photo-acclimation; Kitchen & Zaneveld, 1990; Figure 7b).

Increased concentrations of phytoplankton have also been observed in regions of surface convergence and downwelling. This can arise when phytoplankton with depth-keeping behavior, such as floating or swimming, can counteract downwelling (Lévy et al., 2018; Omand et al., 2011; Taylor, 2018; Figure 7c) or due to the co-occurrence of phytoplankton subduction and lateral advection supplying high chl-a waters into the convergent flow (Figure 7d). In the latter case, the accumulation of phytoplankton will cease and the remaining phytoplankton will be subducted or advected. Enhanced downwelling at submesoscale fronts, dense filaments, and eddies can export phytoplankton away from light (Mahadevan, 2016; McWilliams et al., 2009; Figures 7d and 7e). While some phytoplankton can sustain photosynthesis under very low light conditions, decreased light conditions at depth can still impact net primary productivity (Hoppe et al., 2024; Yang et al., 2021). Downwelling can also export carbon that may eventually become sequestered on climate-relevant timescales (Freilich et al., 2024; Omand et al., 2015). This is accompanied by the upwelling of nutrients on the light side of the front, which can spur phytoplankton growth at the surface (Mahadevan, 2016). In addition to vertical velocities, phytoplankton ecosystems transform rapidly at sub-mesoscales due to lateral mixing and stirring (Martin, 2003; Figure 7f), mixed layer restratification (Boccaletti et al., 2007; Fox-Kemper & Ferrari, 2008; Fox-Kemper et al., 2008), submesoscale instabilities (Mahadevan, 2016), and biological processes such as grazing by zooplankton (Lévy et al., 2018).

Instrument differences may also impact co-located biophysical distributions. Ocean color sensors integrate measurements over approximately one optical depth. The optical depth of the water column estimates the depth at which light penetrates into the water column (Gordon & McCluney, 1975). SST, on the other hand, is derived from the infrared radiation emitted from the thin (<1 mm) thermal skin layer (Hanafin & Minnett, 2002). When fronts are tilted, the ocean color sensor will detect the front when the chl-a change is within one optical depth, whereas the front will appear in SST measurements where the isopycnals outcrop (Figures 7g and 7h). We hypothesize that this phenomena could affect the location and sharpness of chl-a and SST fronts observed by remote sensing (Figure 7g).

Five regions of interest along A2 were selected to highlight fine-scale features observed by MASS (F1-F5 in Figure 6). Chl-a and SST were re-gridded to common 10 m × 10 m grids, and the coordinate system was rotated into the along-track direction. In the following sections, we will hypothesize on the potential mechanisms responsible for the observed physical and bio-optical distributions, but more work is needed to verify mechanisms.

3.4.1. Submesoscale Frontogenesis

F1 is a sharp SST front associated with a forward kinetic energy cascade of around 2×10^{-6} m^2 s^{-3} to smaller scales (Figures 6b and 6d) and peaks of vorticity, convergence, and strain on the dense (warm side) of the front (Figures 8a and 8d–8f). A gradual chl-a front at ≈17.5–17.9 km along the track may be associated with the sharp SST front, but offset due to the mechanisms described in Figures 7g and 7h. The chl-a front is also associated with an increase in PC4 and decrease in PC5. Meter-scale figures on the cold side of F1 (blue dots at y ≈ 500 m between 17.5 and 18.5 km along track; Figure 8a) denote marine mammals breaking up a diurnal warm layer to breathe.

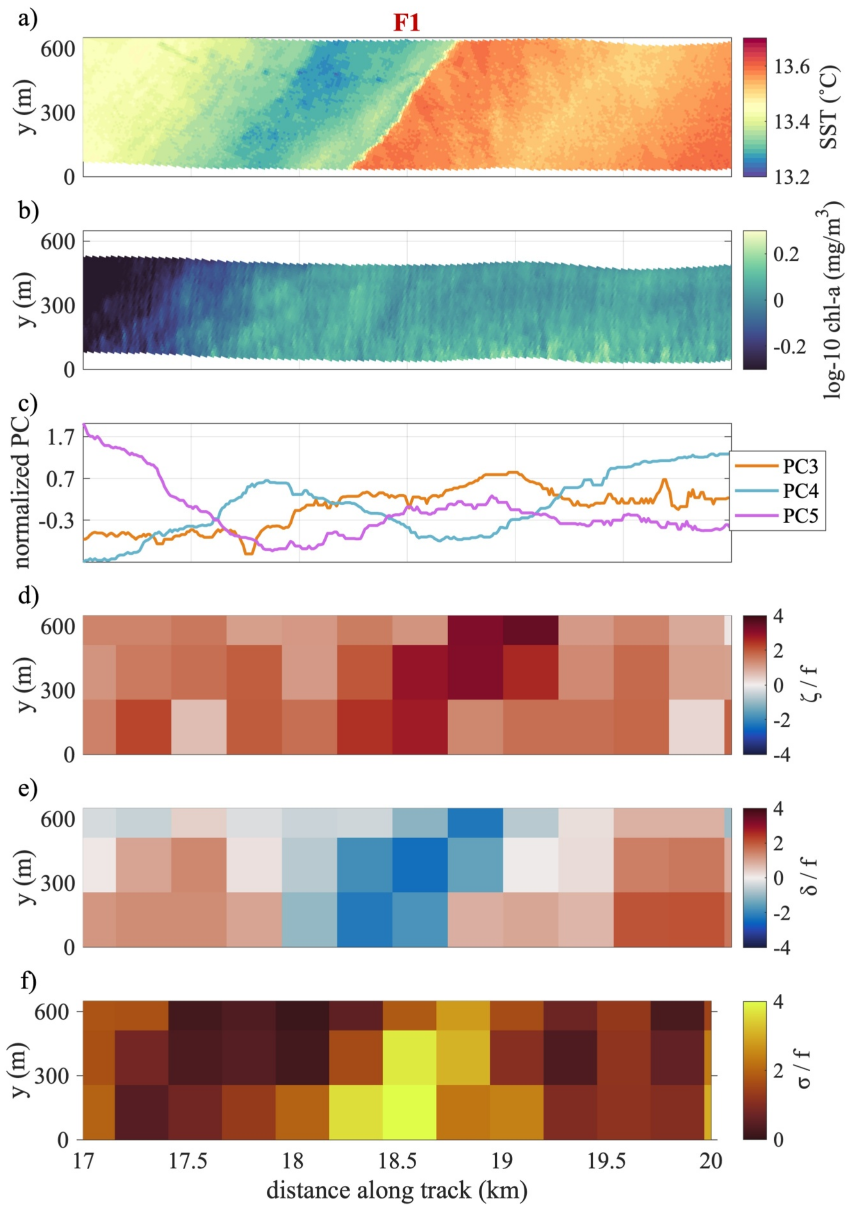

Figure 8. Full swath SST (a), full swath log-10 chlorophyll-a (b), 1-D PCs 3–5 (c), full swath vorticity (d), full swath divergence (e), and full swath strain (f) plotted against distance along track for first feature of interest, F1. Distance across track plotted on *y*-axis as *y* (m). PC's normalized to have a mean of 0 and a standard deviation of 1. Vorticity, divergence, and strain normalized by Coriolis parameter. Central line time was 20:36 UTC.

Current derivatives at F1 (Figures 8d–8f) indicated active submesoscale frontogenesis, which is associated with an ageostrophic secondary circulation in the vertical which upwells water on the light side of the front and downwells water on the dense side of the front (Mahadevan, 2016; Figure 7e). PC4 decreases in the region of convergence (negative divergence; 18–19 km in Figure 8c), which may indicate the downwelling of a phytoplankton group undetected by chl-a. For example, picoplankton, such as prochlorococcus and synechococcus, are abundant in low chl-a waters. Changes in their concentrations may be detected by accessory pigment changes even if not detected in bulk chl-a measurements. Furthermore, phytoplankton community shifts may not be detected if there are concurrent increases and decreases in different phytoplankton groups that yield a near net zero change in chl-a. However, further work is needed to evaluate whether ocean color changes are due to phytoplankton community or physiological changes.

3.4.2. Modulation of a Diurnal Warm Layer by Internal Waves

F2 is a patch of warm SST and high chl-a (highest chl-a between 21.5 and 22.5 km) with a peak in PC3 (~22.5 km Figures 9a–9c). This patch contained meter-scale, warm filamentous features between 22 and 23.5 km, which are smaller than the scale captured by velocity data (Figure 9g). Chl-a and SST anomalies were calculated by subtracting background chl-a and SST fields, calculated from 300 m × 300 m Gaussian smooths, from total chl-a and SST. Warm SST anomalies were associated with positive chl-a anomalies and cold SST anomalies were associated with negative chl-a anomalies (Figure 9g).

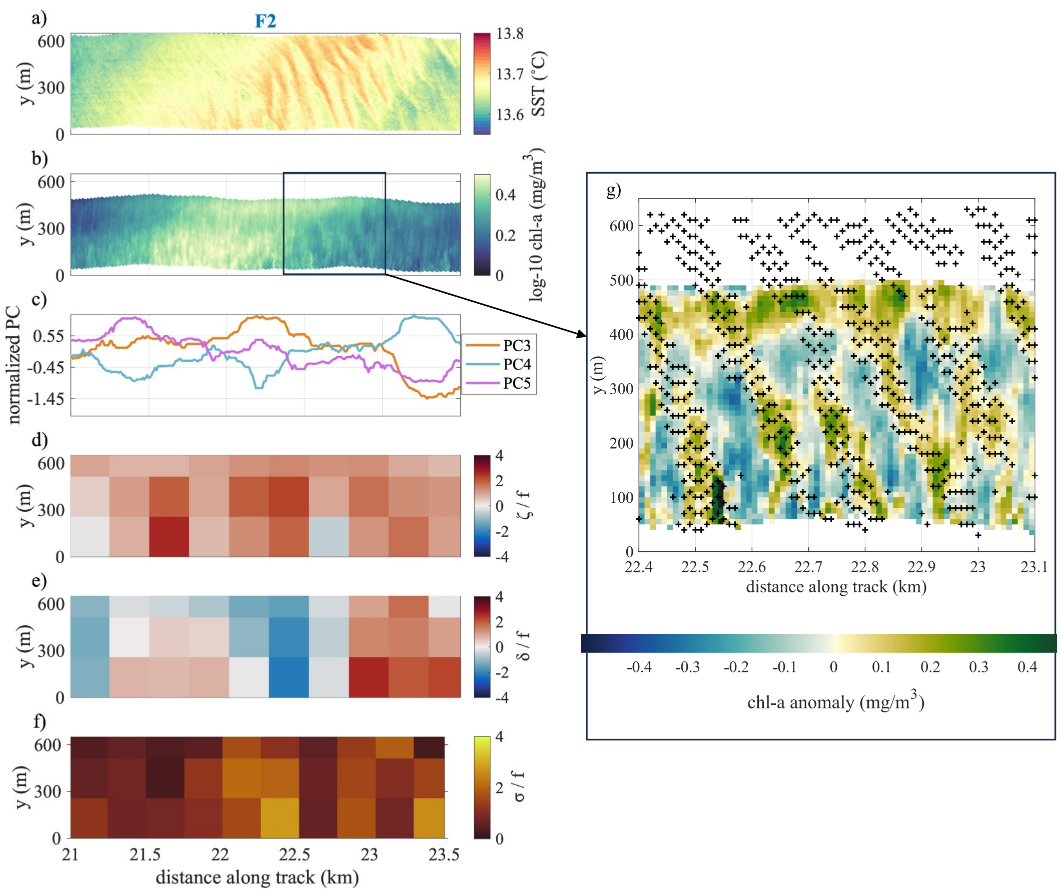

Figure 9. SST, log-10 chlorophyll-a, PCs 3–5, vorticity, divergence, and strain as in Figure 8 for F2 (a–f). Chl-a anomaly between 22.4 and 23.1 km, at location of colocated chl-a and SST "finger-like" structures. Location of positive SST anomaly denoted by + overlain (g). Central line time was 20:33 UTC.

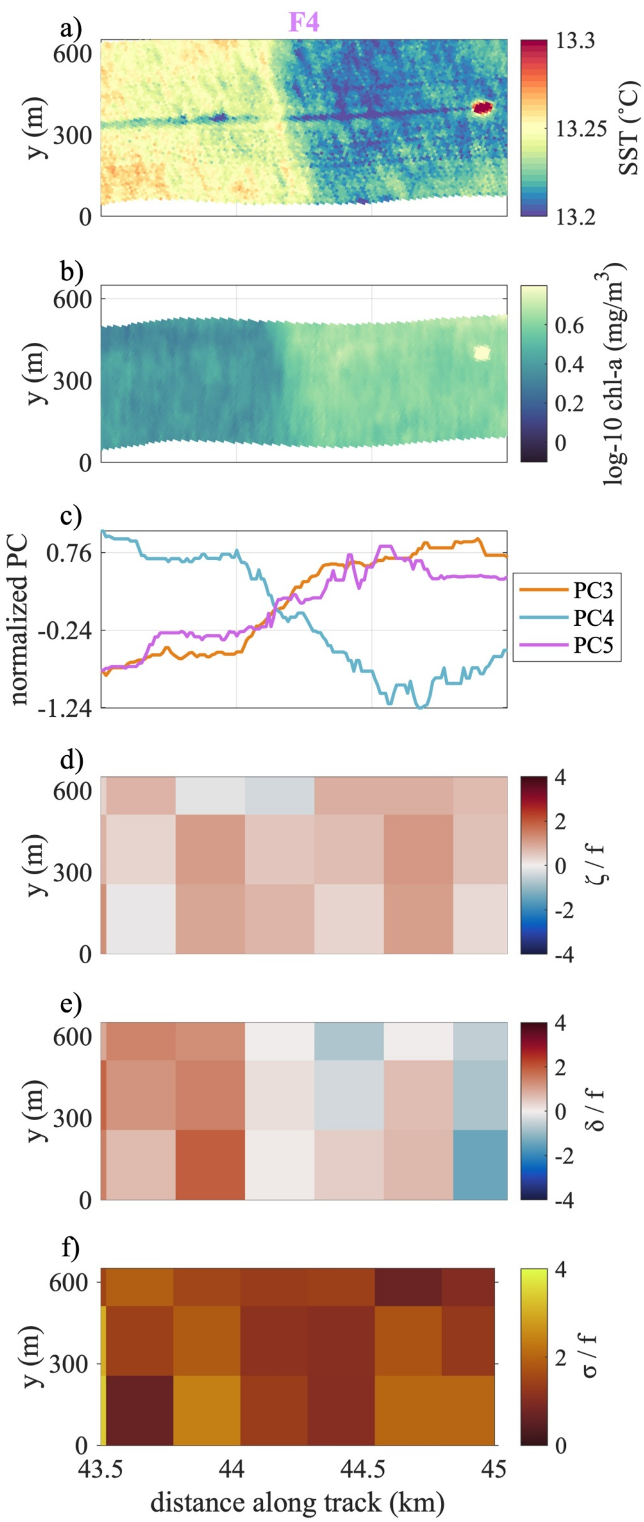

Figure 10. SST, log-10 chlorophyll-a, PCs 3–5, vorticity, divergence, and strain as in Figure 8 for F4. Central line time was 20:26 UTC.

F2 is not associated with a coherent signature in vorticity, divergence, or strain (Figures 9d–9f), so its overall distribution is likely explained by stirring of warm water with high phytoplankton (Figure 7f). F2 also exhibits meter-scale, "finger-like" features marked by warm, high chl-a waters (Figure 9g). This "quasi-periodic" SST variability may be due to the modulation of the diurnal warm layer's depth by internal waves when wind speeds are low and a diurnal warm layer is present (Farrar et al., 2007; Walsh et al., 1998). Patterns of phytoplankton may be driven by its accumulation in internal wave-driven convergence zones (Lenain & Pizzo, 2021; Omand et al., 2011).

F4 provides evidence for a diurnal warm layer. The R/V Oceanus, misidentified in the SST and chl-a images as a region of high SST and chl-a (44.9 km along track), left a streak of cold water in its wake due to the mixing of a shallow diurnal warm layer (Figure 10a). There is no wake in chl-a, suggesting that the chl-a concentration is homogeneous over the depth range mixed by the ship (Figure 10b) while SST, which denotes the skin temperature, is not. Wind speeds are also low, favorable for the development of diurnal warm layers (4–6 m/s; Figure S4 in Supporting Information S1). F4 is a SST and chl-a front with high chl-a, increased PC3 and PC5, and decreased PC4 on the cold side of the front and no strong dynamical signatures (Figure 10). The spatial offset of the chl-a and SST front below $y \approx 200$ m could be due to the differences in measurement integration depths between chl-a and SST (Figures 7g and 7h).

3.4.3. Divergence-Driven Processes

F3 is a patch of warm SST, high chl-a, decreased PC4, and increased PC3 and PC5 in a region of strong negative divergence (i.e., convergence; Figures 11a–11c and 11e). F3 could be associated with mechanisms depicted in Figure 7c or Figure 7d. Decreases in PC4 could be due to the downwelling of phytoplankton, and increases in PC3 and PC5 could be due to the convergence and accumulation of phytoplankton. Whereas F3 displays a high chl-a patch in a region of strong convergence, F5 is a patch of warm, high chl-a water in a region of surface divergence and likely upwelling (Figures 11g–11h and 11k). F5 is also associated with peaks in PC3 and PC5 and a dip in PC4, perhaps pointing to the accumulation of phytoplankton associated with PC3 and PC5. F5 may reflect the impacts of mechanisms depicted in Figure 7a or Figure 7b. In situ phytoplankton pigment data are needed to connect PC's with phytoplankton community groups.

3.4.4. Connecting Mechanisms to Observations

Overall, ascribing mechanisms to relationships observed between hyperspectral ocean color, SST, and surface currents will require a synthesis of numerical and observational methods including process modeling and expanding the spatiotemporal scope of MASS data collection. The impacts of submesoscale dynamics and their vertical velocities on phytoplankton depend on the underlying vertical nutrient and chl-a structures, the depth that vertical velocities penetrate, and the limiting growth factor of different phytoplankton taxa (Mahadevan, 2016). Hence, more work is needed to connect the surface to the underlying vertical velocity, nutrient, and chlorophyll-a structures. Observationally, this can be accomplished by using vertical profilers on ships and autonomous platforms integrating an Acoustic Doppler Current Profiler (ADCP), chlorophyll-a fluorometer to estimate phytoplankton concentrations, backscatter meter to estimate particle load and POC, a sensor to estimate nitrate concentrations (the limiting macronutrient in the CCS), and a photosynthetically active radiation

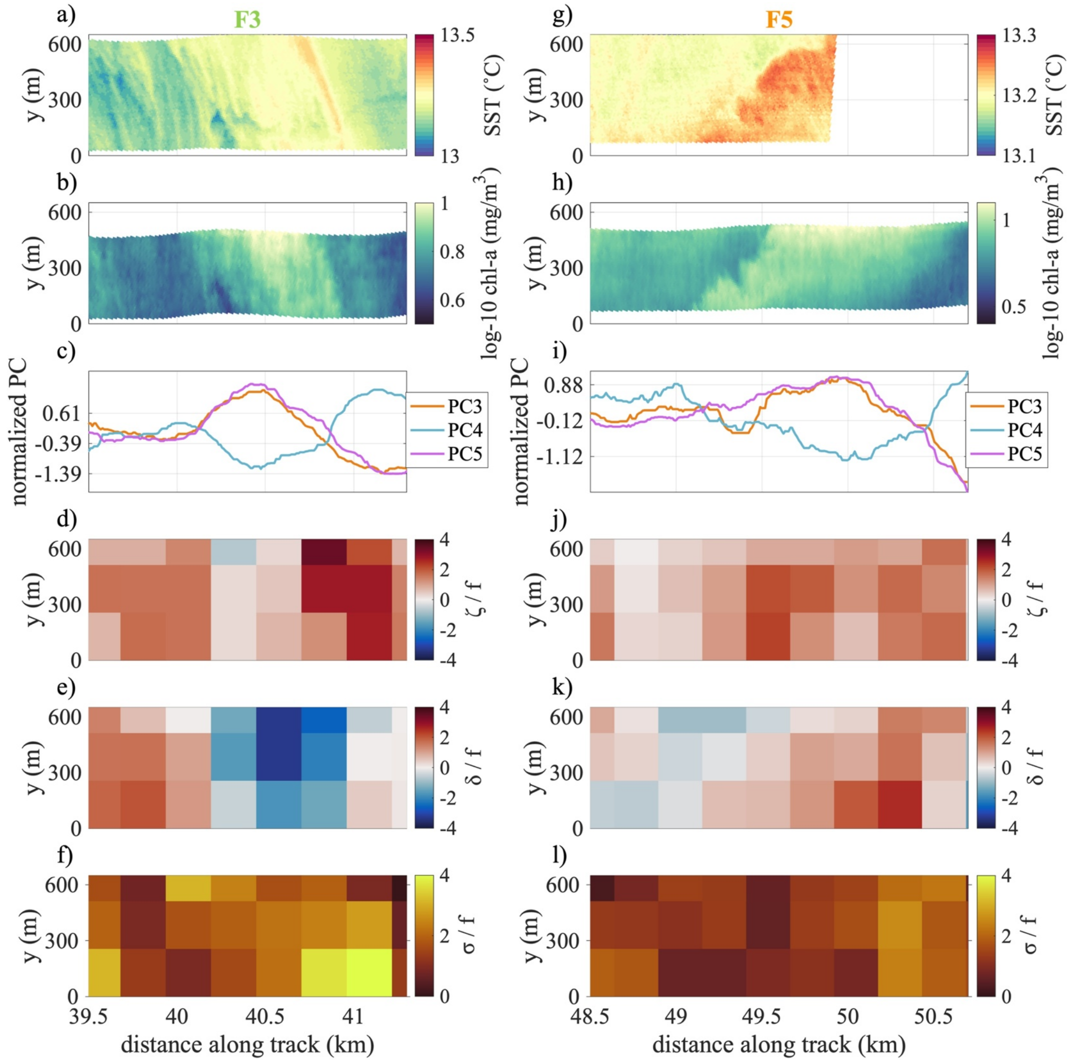

Figure 11. SST, log-10 chlorophyll-a, PCs 3–5, vorticity, divergence, and strain as in Figure 8 for F3 (a–f). Central line time was 20:28 UTC. SST, log-10 chlorophyll-a, PCs 3–5, vorticity, divergence, and strain as in Figure 8 for F5 (g–l). Central line time was 20:25 UTC. SST is cut off due to camera acquisition failure.

sensor. On-board phytoplankton incubation experiments could be used to study the impact of changing nutrient or light conditions on the phytoplankton community structure. Process models will also be useful for describing the physics underlying remote sensing distributions and characterizing the evolution of the horizontal and vertical fields. Characterizing the dynamical mechanisms underlying ocean color observations could also improve the interpretation of paired SWOT and PACE data. Although measuring larger spatial scales than MASS, SWOT resolves some submesoscale ocean processes which can impact phytoplankton community distributions observed by PACE (1 km spatial resolution).

3.5. Ocean Color Methods

The goal of this study is to showcase the potential of MASS for studying submesoscale bio-physical interactions due to its ability to capture concurrent, meaningful information about ocean physics and ocean color. Refining ocean color calibrations and corrections to increase confidence in airborne phytoplankton estimates will be a crucial next step in expanding the scope of hyperspectral airborne data. First, future campaigns should perform an optical calibration of the hyperspectral sensors. This can be achieved by flying over a target of known reflectance. Spectra from hyperspectral airborne platforms should also be compared to in situ radiometry throughout the experiment. Optical sensors are affected by temperature and can experience instrument drift. Spectral calibrations

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should also be performed on both the upwelling radiance and solar irradiance data (Gao et al., 2004). Solar irradiance data from FODIS should be compared to modeled irradiance data under different atmospheric conditions.

Removing sun glint and developing a more robust cloud mask for MASS will further improve the ability to ascribe surface biology changes to submesoscale processes. The best way to avoid sun glint is during data collection, such as by flying away from the glint on the sea surface (Mustard et al., 2001; Wang & Bailey, 2001), which depends on solar position and wind direction. Flights can also avoid windy conditions (Gordon & Wang, 1992, 1994) and fly when the solar zenith angle is between 40 and 55°(Mustard et al., 2001). However, low winds are not optimal for the remote sensing of currents inferred from optical observations of surface waves. A post hoc sun glint correction and filtering of flights with optimal ocean color and physical data, as was done in this study, is most suitable for this type of data set. A sophisticated glint correction could improve agreement between repeat overpasses and ship data. Cloud shadows are another source of artifacts in hyperspectral imagery that should be detected and removed in subsequent studies (Alvera-Azcárate et al., 2021).

Changing viewing and solar geometries between overpasses also presents a challenge for airborne remote sensing. Here, we partially account for changing geometries using statistical methods to predict chl-a and POC from all available flights that made parallel overlaps with the ship track. However, our chl-a and POC models were created specifically for these data. Any subsequent studies should attempt a calibration and in situ evaluation similar to the methods described in Section 2.3. Additionally, more work is needed to test if the model can be applied outside of training conditions. When the model (developed with nadir $r_{rs}(\lambda)$) was applied to every pixel across the full swath, sun glint likely became more significant and raised the values of chl-a and POC plotted in Figure 4. The models also have larger errors at high chl-a values as shown in Figure 3, as typical with log-normal distributions. Lastly, chl-a and POC models were based on band ratios that are not independent. To obtain a POC estimate that is less correlated with chl-a, atmospherically corrected remote sensing reflectances can be used to inversely model backscattering, a proxy for POC (Stramski et al., 1999).

Hyperspectral measurements can yield products beyond chl-a and POC. Principal component analysis applied to the full hyperspectral data decomposed the data into spectral signatures associated with water and its optical constituents. For example, PC1 seemed to capture the variability of particulate backscattering, PC2 captured the variability of chl-a, and higher PC's captured small peaks potentially associated with phytoplankton pigments. Glint resulted in significant across-swath artifacts when the PCA was applied to full swath data. Glint correcting ocean color data will be a crucial step for assessing the across-swath variability in hyperspectral ocean color changes. Additionally, more work is needed to determine if individual PC's capture meaningful variability in phytoplankton accessory pigments and other optically active constituents of the water column. The association between higher order principal components and phytoplankton pigment groups is often tested in principal component regressions (Bracher et al., 2015; Lange et al., 2020). The coherent spectral shapes of the principal components and their spectral features indicate that a principal component regression using MASS data may be able to predict in situ pigments. MASS, once atmospherically corrected, could be used in conjunction with algorithms that rely on resolving narrow spectral features, such as those related to accessory pigments (Chase et al., 2017; Kramer et al., 2022). In situ validation data of pigments or phytoplankton community composition can be used can be used to evaluate or regionally tune algorithms predicting phytoplankton community groups (Kramer et al., 2022; Lange et al., 2020). Future studies combining MASS currents and SST with phytoplankton community distributions could help disentangle the impact of submesoscale dynamics on phytoplankton community groups.

4. Conclusion

This study highlights the potential for suborbital remote sensing to study the impact of submesoscale processes on phytoplankton ecosystems and carbon transport. Merging bio-optical and physical airborne remote sensing data from the same platform avoids the spatiotemporal aliasing impacting in situ sensors and co-located remote sensing observations from different platforms. Here, we conducted a regression analysis to derive chlorophyll-a and particulate organic carbon from low altitude ocean color data, without first correcting for atmospheric effects. We merge concurrent airborne snapshots of sub-kilometer ocean velocities and their derivatives (i.e., vorticity, divergence, and strain), ocean color, and sea surface temperature to show examples of potential impacts of submesoscale dynamics on phytoplankton. This study works toward the detection and quantification of submesoscale bio-physical mechanisms using remote sensing. This could lead to the better understanding of the impacts of submesoscale dynamics on ocean biogeochemistry and their contributions to the ocean's role in climate.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Presented data and code to generate figures were archived with the UC San Diego Library Digital Collections (https://doi.org/10.6075/J0RN386Z). Data in this paper was collected during the Submesoscale Ocean Dynamics Experiment (S-MODE). The S-MODE data presented in this manuscript is publicly available via the Physical Oceanography Distributed Active Archive Center (PO.DAAC) at https://podaac.jpl.nasa.gov/S-MODE. A full description of in situ data collection and bio-optical calibrations can be found here, in the SMODE Bio-optical Data Report for Pilot Field Campaign (Lang et al., 2023): https://archive.podaac.earthdata.nasa.gov/podaac-ops-cumulus-docs/s-mode/docs/SMODE_PILOT_Bio-optical_Cals_V3.pdf. Hyperspectral data from the Modular Aerial Sensing System (MASS) can be downloaded here: https://doi.org/10.5067/SMODE-MASS1H (S-MODE Team, 2022a). MASS long wave infrared imagery (LWIR) can be downloaded here: https://doi.org/10.5067/SMODE-MASS1I (Lenain & Statom, 2023). MASS DoppVis imagery can be downloaded here: https://doi.org/10.5067/SMODE-MASS1D (Lenain & Statom, 2022). Shipboard bottle data can be downloaded here: https://doi.org/10.5067/SMODE-RVBOT (Lang et al., 2022). Shipboard flow-through data can be downloaded here: https://doi.org/10.5067/SMODE-RVTSG (S-MODE Team, 2022b). Sentinel-3 OLCI Level-2 (Full Resolution) and Sentinel-3 SLSTR Level-2 Sea Surface Temperature (SST) data can be downloaded from the EUMETSAT User Portal here: https://data.eumetsat.int/. Data presented in Figure 2 was captured on 29 October 2021 in the following bounding box: (36.1, −126), (36.1, −121), (38.3, −126), (38.3, −121). Solar zenith and azimuth angles were calculated using a Solar Position Calculator in MATLAB (Mikofski, 2024).

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